

PROJECT DELIVERABLE 4

Netflix Content Analysis: Project Recap & Lessons Learned

AIT-664-DL3: Information Processing, Representation and Visualization

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Dr Charles Lynch

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TEAM MEMBERS

Nithiya Varsha Markandan Rajendren- G01450937

Mohan Silambarasu Elangkumaran- G01524739

Sonal Kumar Masid - G01514651

Introduction:

Netflix, being one of the leaders in entertainment industry has revolutionized how consumers and audience view movies and tv shows. Over the years, Netflix has been primarily focusing on international markets and into content production. In this project we focused on analyzing how Netflix's content strategy has changed over the time. Our main goal is to identify various trends and study the changes in distributions of content types, popularity in genre and production based on each country. We also explored various rating and release trend characteristics which provides more insights about how Netflix strategized these contents in order to meet the preferences of viewers across the world.

Initial Requirements:

In this project, we wanted to analyze how Netflix's content strategy evolved over time, mainly focussing on certain areas like how content types like Movies and TV shows have evolved over time, identify various trends in genre popularity and which genres do viewers prefer now. We also aimed to analyze the trends in international content production by Netflix and also understanding the patterns in rating distributions and content duration.

We believe these outcomes would help Analysts, strategists to better understand Netflix's strategy over the last decade and patterns of viewer consuming Netflix. In order to achieve these outcomes, we needed structured and well defined data inputs such as Content Type, Release Year, Date Added , Rating, Genre and Duration. We also made sure we had all types of data including categorical and numerical variables.

Research Question:

We then focused on forming key research questions before working on analyzing, which are show below:

Our main research question is How has Netflix's content strategy evolved over time in terms of content type, genre popularity, international production, content ratings, and duration characteristics?

This research question aims to find out how a big entertaining platform like Netflix has used the content over the years which provides a solid foundation in the company's revenue and future growth. This research question is a very strong as it is clear, focused on certain key areas as mentioned in the research question which is feasible at the same time involves complex analysis of Netflix's data.

Hypothesis:

We had written a long hypothesis answering all the research question individually, later based on the feedbacks that were given to us , we have revised our hypothesis such that it provides more clarity and it is very well aligned with the research question. The final Hypothesis is:

There is a significant increase in the percentage of TV shows relative to movies across the globe over the past decade, with a growing focus on content rated for teen and mature audiences.

Data Acquisition:

We have sourced the dataset [1] from Kaggle. Kaggle is one of the most reliable source for the dataset as these datasets are not only reliable but also validated, structured which is why many of us from the data science community use this. The selected dataset initially contained over 6,234 records where it also included the metadata of Netflix, it also contained information of variables like type, title, date_added, release_year, country, genre, rating, and duration which aligns very closely without research analysis.

This dataset included all type of variables like categorical, numerical and even textual for a complete analysis. Variables like type, country, rating, Listed_in are categorical and Variables like Release_year, Duration were numerical ones whereas title, description, cast, director were mostly text-based ones. Hence, this dataset is very well structured and it includes all required data characteristics such as defined, measurable, unitized, relatable, normalized and quality.

Data Preparation and Cleaning:

Although the dataset was well structured, we need to perform some data cleaning to make the dataset more suitable for our analysis. We initially worked on handling missing values such as certain text based variables like director, cast and country were replaced with unknown in records where they were empty or missing. We also worked on formatting date_added columns and We also had to separate columns for duration in movies and TV shows as both were initially present under one column. We then replaced all the null and missing values in rating column with Not Rated. We also worked on mapping standard country names using PyCountry library and removed duplicates using df.duplicates() method. These steps made sure the dataset is more measurable, normalized and aligns well with our research question.

Information Visualization (Data Analysis and Visualization):

Now, after working of collecting, preparing data and then cleaning the dataset to make it more relatable to our analysis, we then started with the main focus of our research which is Exploratory Data analysis of our Netflix content dataset.

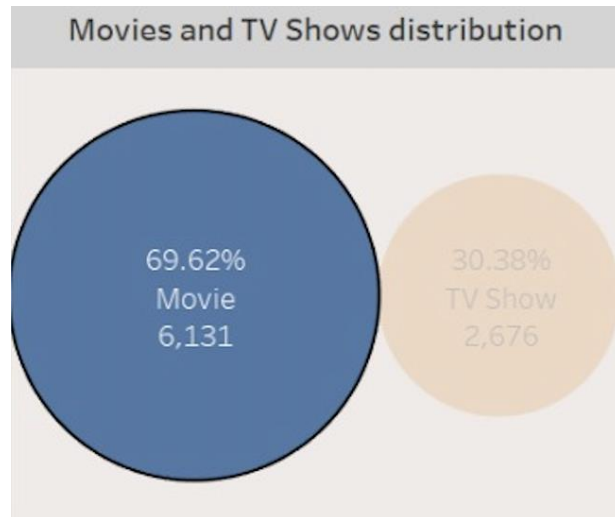
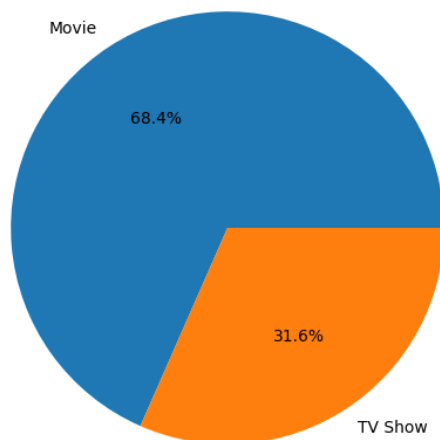
Content Distribution: Movies and TV shows

Initially we started with the exploratory analysis by studying how two of the content types such as movies and TV shows are distributed in the content catalog. Here in this analysis

we used both python and Tableau for content distribution. We used both python charts and time series line charts to generate insights about content distribution.

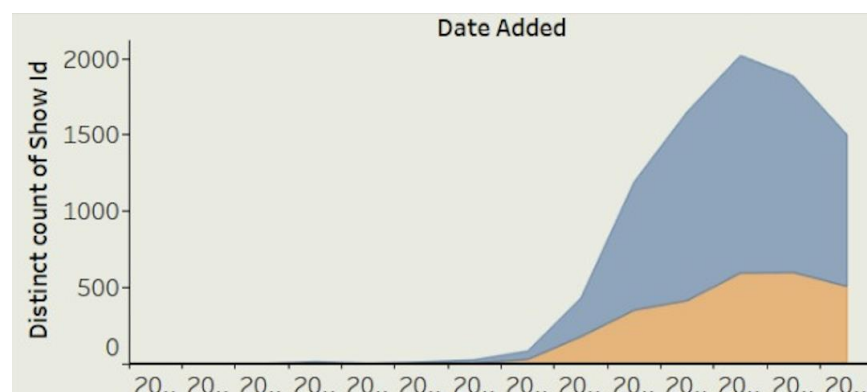
From the below Pie chart and Donut Chart, it is evident that nearly 68.4 percent of the Netflix content consists of movies where only 31.6 percent where from TV series in spite of the demand they hold.

Distribution of Content Type on Netflix



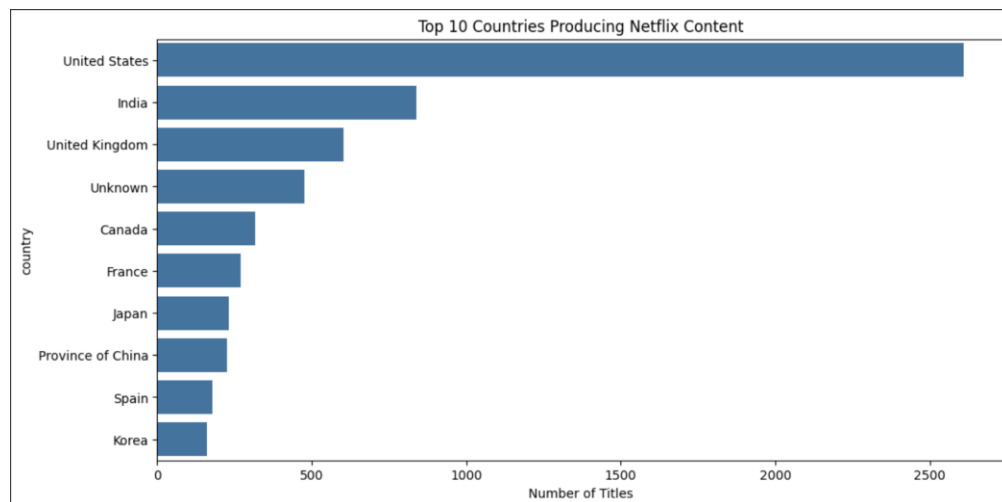
The same has been represented in tableau as well using donut chart , Out of those 8,807 initial entries only 30.38 percent were all TV shows and the remaining majority of 69.62 percent were still composed of movies.

We also used tableau for exploring time series analysis , where we could see from 2008 to 2021 , Netflix did increase a number content every year. However, the increase peaked only around during the covid 19 pandemic era between 2019 and 2020. Also TV shows have shown some great increase after 2015 mostly peaking around covid time again. This supports our hypothesis on how Netflix has shifted their focus to TV shows to gain more user views and generate more revenue in recent years.

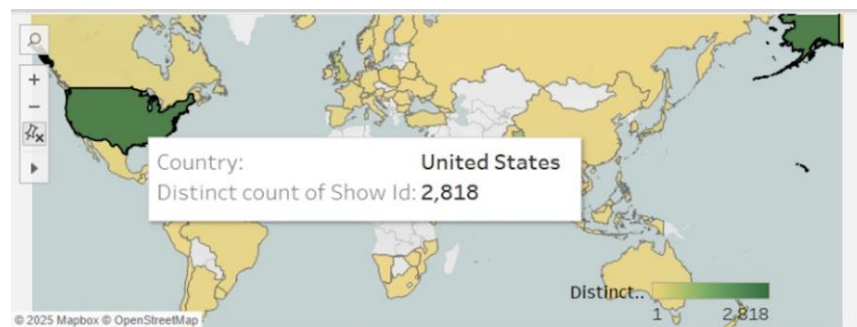


Country wise Content Production:

In this country wise content analysis ,we used bar graph as a visualization technique using seaborn and plotted the countries that produce content on world map as well. From the below bar graph, it is clear that United States are the ones which produce more Netflix content over 2500 contents which is then followed by India, Uk and Canada. These countries play a significant role in entertainment industry especially in platforms like Netflix.

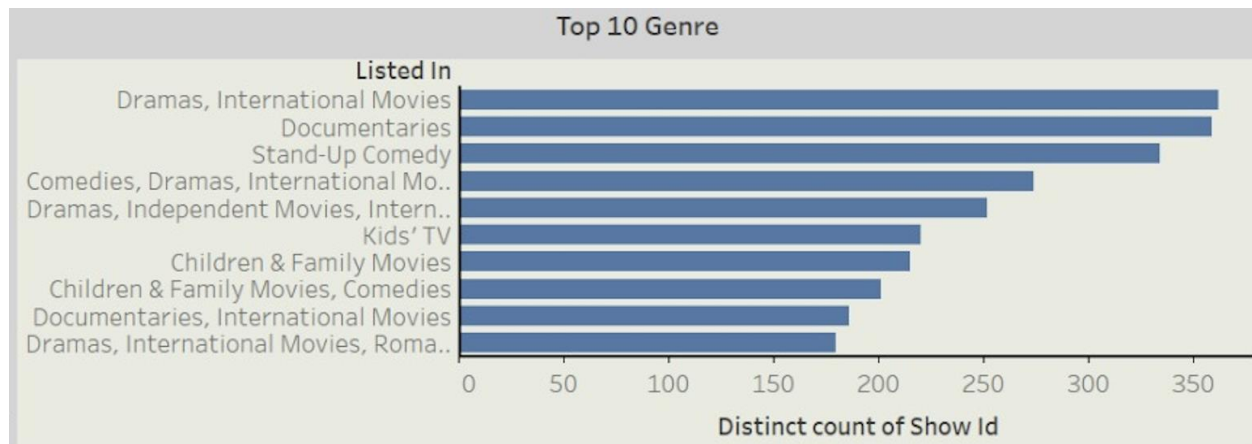


We also performed a similar analysis on tableau as well, This also showed that there are contents being produced from countries which are not English speaking countries. This also proves our initially assumption if Netflix has has expanded production across the globe after pandemic especially due to popular shows like Money heist and Squid game.



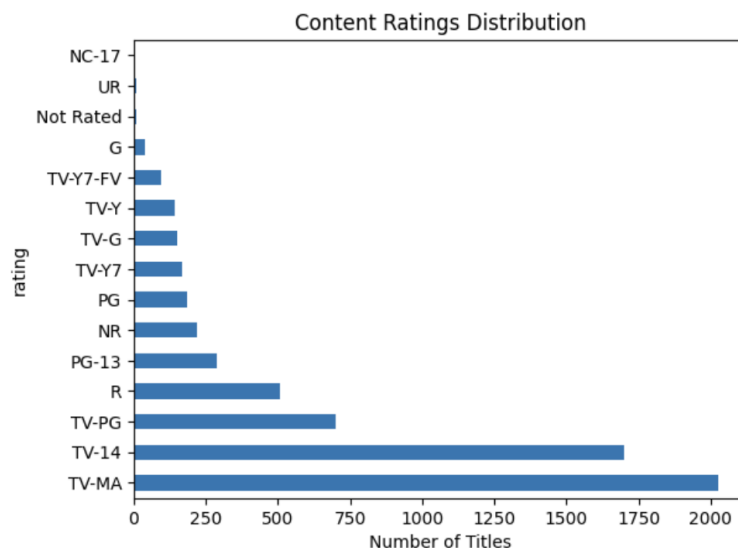
Genre Popularity Trends:

In this visualization we used bar chart using Tableau to showcase which genres are the most popular ones. From the below graph it is evident that the most common and favourite genres being Dramas, International Movies, and Documentaries which actually confirms our initial hypothesis that documentaries are more popular now a days when compared with rom com shows.



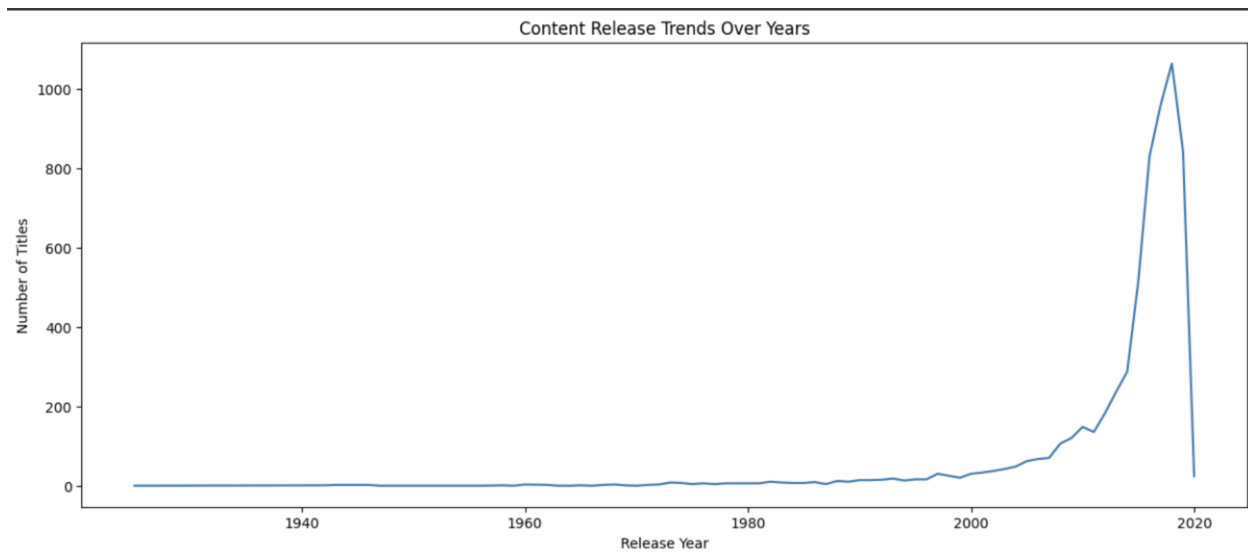
Distribution of Ratings:

Here, we used bar graph using python , Tableau for rating distribution analysis as bar graph would give a clear picture on how various ratings categories are distributed over the content. From the below graph it is evident that the two most categories are TV-MA and TV-14 which shows a large section of viewers of these platforms are mature and teen audiences. But Surprisingly, TV-MA dominates over teen focused TV-14 contents in terms of ratings distribution.

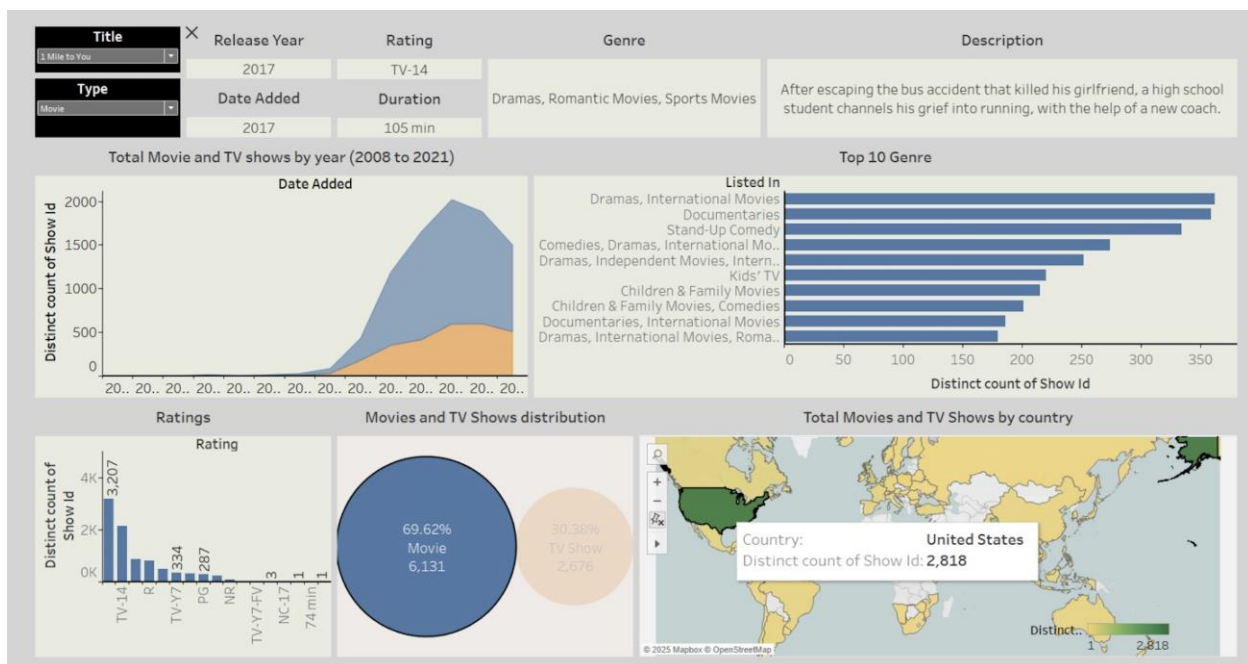


Content Release Trend:

In this analysis, we focused on how Netflix releases content over the years in order to keep the viewers engaged throughout these years. We used a simple line graph to specifically shows the highs and low points in this trend analysis. It is clear from the below line graph that, the release of content has increased over the years peaking after 2015 with the most number of content releases occurred during 2019 pandemic.



The below snippet shows the dashboard of Tableau , which showcases all the analysis and visualization we performed throughout our analysis . We have covered various visualizations right from time series chart, bar chart, donut chart and even world map distribution. These visualizations provide a multi-dimensional view on various patterns and trends in Netflix's content strategy.



Information Modelling:

After performing Exploratory Data analysis, we then focus on topic modelling in our modelling phase. We applied some in depth machine learning techniques to identify major themes to explore about deeper relationships and outcomes.

We explored Topic modelling which is a type of unsupervised machine learning technique which is used to classify and predict the themes in the given texts. We used the Latent Dirichlet Allocation for this topic modelling. In this method, we first applied CountVectorizer to convert the text into numerical text tokens. After that we filtered top 100 frequent terms, we made sure to not include English stop words. Each topic is then distributed over words and then we trained the LDA model with to extract top 10 words from each topic.

```
Topic 1:
young man woman help police new finds women father son
Topic 2:
life love year family woman old young man true town
Topic 3:
documentary series world secret life war power evil follows team
Topic 4:
new comedy special stand world comedian life high school friends
Topic 5:
friends new years school home family teen life group girl
```

The above screenshot shows the five topics that are being extracted. Here the topic 1 shows that are related to thriller, dramas, where topic 2 is mostly about relationships , love stories, and topic 3 is about documentaries. The topic 4 is surprisingly shows comedy genres mostly tied with youth type humors and the final topic is related to the usual teen dramas keywords.

We could see topics like Family dramas, thrillers, youth humor, romantic stories were extracted. These align very well with the genre popularity we performed during analysis phase. We then moved to Linear regression, to predict the count of Netflix movies based on the given certain parameters in the dataset like ratings and release year. Initially we transformed the categorical variable using oneHot encoding. Then we split data into training and testing data sets. A linear regression model was trained, and the model's performance was calculated using a mean square value to evaluated the error in prediction.

Effectiveness of Analysis:

In this part of our research, we explored a wide range of data analysis techniques to get some insights before moving on to modelling. We worked on exploring methods like content distribution analysis, country wise content production, Genre Popularity trends , Rating distribution, Content Release trends and outlier detection.

We used various visual models like Pie chart and Donut chart to showcase distribution of content types, using both Tableau and python. We also used bar graphs for country wise content production and rating distributions. We also used Line graphs as a visual model to identify the peaks and lows of the release trends of various contents.

From this research analysis, we could say we have addressed our research questions and we could see most of our hypothesis were aligned with our analysis .

Our analysis results as discussed in the previous section proved that TV shows have increased over the recent years. The analysis also confirmed that there is a growing number of international content through our bar graph and world map visualizations. This indicates our initial analysis of growth of international content in non-English-speaking countries like money heists and squid game. The only surprise we had was regarding TV-MA content being dominant over TV-14 which shows there are more shows and movies for the mature audience rather than teen shows. We performed a systematic approach right from collecting data, preparing and cleaning data to make it analysis ready dataset. From this structured approach, we could see the detailed analysis and deeper visualization have a great impact on generating useful and interesting insights supporting our hypothesis.

Challenges Faced:

We had initially created a very broad hypothesis and as per the professor's feedback we had revised our hypothesis to make it more research aligned and provide more clarity. Our revised hypothesis supports our in detailed analysis and visualization showcase the importance of choosing the right hypothesis for the research.

Initially we faced few struggles while working on preparing dataset to make it analysis friendly. We had to clean up the data replacing all the missing values in multiple columns like director , cast and country with Unknown and we had to replace the ratings column as well. We also had to format columns like date_added, split duration columns and also mapped country with standard ISO names using PyCountry library. These extra steps helped in providing more reliable analysis for our research.

We also received feedbacks on providing the citations for the tools and dataset which showed how important is to cite the appropriate tools and to follow proper formatting while writing reports.

Key Learnings:

From this research project, We learnt how to use a real life dataset and follow various steps right from cleaning, preparing to visualizing and modelling which has a real world application. There were multiple key learning that we learnt from this project either from professor's feedbacks or while working on analysis.

We realized it is highly important to focus on a more singular version of hypothesis rather the hypothesis being in multiple parts. This helped us to narrow down what to focus on during our research. We also learnt proper citations are very important for standard reports.

We have explored various new tools and techniques like tableau , Linear regression , Modelling and visualization techniques while solving the Netflix content strategy , this have opened us to new technologies which would greatly impact in our data analytics pathway.

Improvements:

We have performed most of the analysis as per our plan , following our hypothesis. We have explored various Content strategies put forth by Netflix using data analytical tools on a structured dataset, However in future we would like to focus on more larger and unstructured datasets. Exploring such unstructured would be even more challenging to clean and analyze the dataset. We would also like to perform more deeper modelling using models like clustering or perform any sentiment analysis for the textual data to discover more deeper insights on what keywords are more popular across the globe.

Tool Usage and Justification:

In this Netflix Content Analysis, we have used a variety of tools and libraires to complete each phase in our project. We began by sourcing our dataset from Kaggle which was ideal for our analysis. During our initial preprocessing, cleaning we used Python libraries such as NumPy, Pandas for numerical code and we used seaborn and matplotlib for generating visualization. For more advance visualization we also used Tableau for building dashboard including time series charts, donut chart and even world map. While modelling we used scikit-learn for applying LDA and linear regression. In the end, these tools helped to provide a strong foundation for this project.

Conclusion:

In this entire research project, we performed an end to end analysis starting from picking the research question, initial hypothesis to data acquisition, cleaning, analysis and visualization. Initially we faced challenges like formulating the right and singular hypothesis and we also faced issues during cleaning of data by handling missing values, formatting certain columns, restructuring duration fields, all these showed us that data analysis is a continuous process which is interlinked to each and other.

We performed informational modeling and visualization using python libraries , tableau to identify patterns in content type distributions , genre popularity , international content production and rating areas. We also used topics modelling and linear regression to discover deeper insights as well.

Our final findings suggest that with help refined hypothesis we were able to narrow down our research and our findings supported the hypothesis. Visualization model showcases these insights in a clearer way. Inspire of early challenges and data imperfections, this project did provide a great technical and analytical experience to us. In the future, our work could expand into data analysis on unstructured dataset and also applying Natural language processing techniques such as sentiment analysis on keywords and even user views, thus providing more deeper insights into viewer preferences and content strategies.

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